Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Lesson strip

Today I can

Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence.

Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples.

Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

In your book

In your book: Glue in today's strip; date it. Answer the four clues.

Use: Amylase and pH: Continuous-Sampling Investigation

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Arrival · Supported

1. Enzyme and substrate

Link to today: Amylase is the enzyme; starch is its substrate. Which one speeds up the breakdown reaction?

An enzyme is a biological catalyst: it speeds up a reaction without being used up.

2. Read the colour

Iodine becomes blue-black with starch. A sampled drop turns blue-black: is starch detected?

Blue-black = starch detected. Orange-brown = starch not detected.

3. Recall binding

Which substrate fits amylase’s active site: starch or an unrelated substrate?

Recall the enzyme and substrate model from the previous lesson.

4. Prepare the evidence

Which comes first: prepare the iodine wells / mix and start timing?

The reaction continues after mixing, so the sampling equipment must already be ready.

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Exit Ticket · Supported

Evidence

These are example results at pH 7: 30 s blue-black; 60 s blue-black; 90 s orange-brown. Blue-black means starch was detected. Orange-brown means no starch was detected in that sample. The planned interval is 30 s. The amounts at the start match. The temperature and the endpoint rule match too.

1. Choose the first observed endpoint time: 60 s or 90 s. What does its colour tell you?

Use the colour key. Endpoint = first sample with starch not detected.

2. A friend writes 75 s exactly. Use the sampled times to help them repair it.

Choose an observed time. Say “We did not observe …”.

Under the heading Exit ticket: answer the short questions in your book.

Use: Amylase and pH: Continuous-Sampling Investigation

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Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Arrival · Independent

1. Enzyme and substrate

Identify the enzyme and substrate in amylase acting on starch.

Enzyme = catalyst. Substrate = molecule the enzyme acts on.

2. Read the colour

At 60 s iodine is blue-black; at 90 s it stays orange-brown. Which sampled time first meets the endpoint?

The agreed endpoint is the first sampled time when starch is not detected.

3. Change and measure

State the independent and dependent variables, then name one condition to keep constant.

Change buffer pH; measure endpoint time. Controls include temperature and starting solution amounts.

4. Prepare the evidence

Why prepare the iodine wells and time headings before mixing?

Once amylase and starch are mixed, the reaction continues while equipment is being prepared.

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

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Exit Ticket · Independent

Evidence

These are example results at pH 7: 30 s blue-black; 60 s blue-black; 90 s orange-brown. Blue-black means starch was detected. Orange-brown means no starch was detected in that sample. The planned interval is 30 s. The amounts at the start match. The temperature and the endpoint rule match too.

1. State the first observed endpoint and the window in which it was reached.

Use “record ___ s; after ___ s and by ___ s”.

2. Give one way to protect a fair comparison across pH values and explain why it matters.

Use one matched condition: temperature, starting amounts or sampling interval.

Under the heading Exit ticket: answer the short questions in your book.

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

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Arrival · Aspire

1. Enzyme and substrate

Why does an iodine colour change provide indirect evidence of amylase activity?

Iodine detects starch in a sample; it does not show the enzyme molecules working.

2. Read the colour

At 60 s starch is detected; at 90 s it is not. Is an exact endpoint of 75 s justified?

No observation was made at 75 s. Distinguish an observed endpoint from its possible timing window.

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

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Arrival · Aspire · continued

3. Change and measure

Distinguish controlled temperature from a comparison with no active amylase.

A controlled variable stays matched. A comparison without active enzyme checks whether the enzyme is needed for the observed change.

4. Prepare the evidence

The 60 s sample is missed. The endpoint is seen at 90 s. Can that endpoint still have the same 30 s timing window?

The last sample that showed starch might now be at 30 s. Use only times that were seen.

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

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Exit Ticket · Aspire

Evidence

These are example results at pH 7: 30 s blue-black; 60 s blue-black; 90 s orange-brown. Blue-black means starch was detected. Orange-brown means no starch was detected in that sample. The planned interval is 30 s. The amounts at the start match. The temperature and the endpoint rule match too.

1. If the 60 s sample was missed, how would your endpoint statement and timing window change?

Use only 30 s positive and 90 s negative; do not invent the missing result.

2. Name one authorised way to improve this timing limit. Name one uncertainty that it does not remove.

Link the improvement to the gap between sample times. Then think about judging colour.

Under the heading Exit ticket: answer the short questions in your book.

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

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Starter and method strip

Starter

Which setup protects the first observations?

Method strip

Get the centre-approved setup ready. Label the iodine wells. Set out the timer. Agree the roles.

Check with the adult that all is ready. Stop and check before mixing.

Mix and start timing together. Follow the authorised method.

Take a separate sample at each agreed fixed time gap. Put it in an iodine well you have set up.

Keep the raw time and colour results. Keep any gaps in the record.

Write what you saw in your book. Write one conclusion about pH and amylase activity. Use the evidence to support it.

Safe access

Keep only the kit for this stage on the table. If you cannot take part safely, watch the teacher show the task. Use the same observation and conclusion.

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

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You Do · Independent

State the first observed endpoint for pH 7 and the interval in which it was reached, using the supplied observations. [2 marks]

Order the four method cards. Explain why preparation must happen before mixing. [2 marks]

State the independent variable, dependent variable and one matched control, explaining why the control matters. [4 marks]

Repair the proposed exact endpoint of 75 s. Explain what should be recorded if a sample is missed. [2 marks]

Compare two pH values using their times. State one limit of the conclusion from this first series. [3 marks]

I · S · C

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

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You Do · Supported

1. Read the pH 7 timeline. Name the first orange-brown sampled time and say what that colour means.

Use the colour key beside the three wells. First means the earliest sampled time meeting the rule.

2. Put the four method cards in order: prepare and check; start together; sample at fixed times; keep the raw observation.

Read each card’s action word. The adult authorises practical steps; sorting is a model of the sequence.

3. Choose one thing to keep matched: temperature / buffer pH. Say why it stays the same.

Choose the condition that was not deliberately changed. Try “Keep ___ matched so we compare pH.”

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

You Do · Supported · continued

4. Repair “endpoint = 75 s exactly” using the observed 60 s and 90 s colours.

The stopwatch was read at sampled times only. Choose 60 s / 75 s / 90 s, then use the colour clue.

5. Compare the pH 7 and pH 4 times. Choose the faster breakdown and give one number as evidence.

For the same starting starch and endpoint, shorter time means faster breakdown. Compare 90 and 240.

I · S · C

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

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You Do · Choice

Choose the first orange-brown time at pH 7: 60 s / 90 s / 120 s. Choose its window: after 60 s and by 90 s / exactly 75 s. Record the time and colour meaning. Record C beside your book response.

Order the shuffled cards: keep observations; sample at fixed times; prepare; start together. Choose why preparation precedes mixing: avoid missed early samples / make colour brighter. Match pH / endpoint time / temperature to change / measure / keep the same; give one control reason.

Choose the truthful missed-sample record: labelled gap / invented 75 s. Compare pH 7 at 90 s and pH 4 at 240 s: choose the faster result and quote a time. Choose the claim: these trials / every amylase. Give one reason.

After choosing, write one short scientific sentence, label or reason in your book.

I · S · C

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

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You Do · Aspire

For pH 7, give the endpoint time we record. Show how this differs from the true endpoint window. State how wide the window is. [3 marks]

To find the window width, subtract the two times when samples were taken. Do not call it an exact ±15 s measurement.

Explain why we get the tile ready first. Explain why we take samples at a fixed time gap. Show how these protect different parts of the evidence. [2 marks]

Think about missing the start. Then think about what we cannot know between samples taken at fixed times. These are two different problems.

Explain why we keep temperature the same. Explain why we use a comparison with no active amylase. Show how their roles differ. [2 marks]

A comparison group and a controlled variable are not the same. For both, match the conditions that matter.

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

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You Do · Aspire · continued

Evaluate this claim: The 60 s sample is missing in a second record; 30 s is blue-black and 90 s orange-brown. State what is now known and repair the claim “the same precision”. [3 marks]

Use actual observations 30 s and 90 s. Their gap is 60 s even though the intended interval was 30 s.

Compare pH 4 and pH 7 with the times given. Keep your claim to these trials. You might expect a smooth trend. A colour might be hard to read. Explain why neither reason lets you change a raw observation. [3 marks]

Keep your claim tied to the conditions of the test. Keep a result even if it is a surprise. This protects the evidence.

I · S · C

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

You Do · shared evidence

Evidence and reference diagrams

Enzyme: biological catalyst; speeds a reaction without being used up. Substrate: substance it acts on. pH: a measure of acidity or alkalinity. Amylase acts on starch, producing smaller sugars.

Iodine key: blue-black = starch detected. Orange-brown remains = starch not detected. check separate sampled drops. Endpoint = first sampled drop leaving iodine orange-brown; record its time / s.

Method card | Action

1 | Prepare and check: approved setup, labelled wells, timer and roles ready.

2 | Start together: mix and start timing as the approved method states.

3 | Sample at fixed times: separate samples into prepared iodine wells.

4 | Keep raw observations: record colour and time honestly as sampling continues.

Supplied illustrative observations — not your practical results. Single runs; equal starting amounts and the same iodine endpoint. Samples every 30 s. pH 7: blue-black at 30/60 s, orange-brown at 90 s. Record 90 s.

pH | First endpoint / s

4 | 240

5 | 180

6 | 120

7 | 90

8 | 150

Use: Amylase and pH: Continuous-Sampling Investigation

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Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

You Do · shared evidence · continued

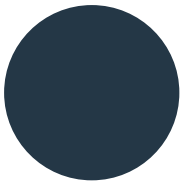
Evidence and reference diagrams · continued

For a valid comparison, change pH. Keep temperature the same. Match enzyme and starch concentrations and volumes. Match the total volume, the start and how you take samples. A separate control with no active amylase should still test positive for starch. That control is not the same as a factor kept constant.

A separate pH 7 repeat gave 120 s. Keep both 90 s and 120 s. A further run had no endpoint by 300 s. Record what was seen. Do not write zero. A paper test uses observations that are given to you. It does not show that you handled reagents.

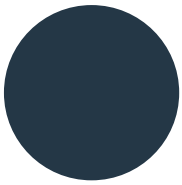
Reference

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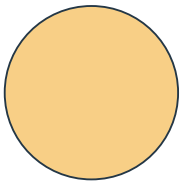
30 s

Blue-black



60 s

Blue-black



90 s

Orange-brown

Supplied example: samples every 30 s.

Use: Amylase and pH: Continuous-Sampling Investigation

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Knowledge organiser

Learning goal

I can use continuous-sampling evidence to investigate how pH affects amylase activity.

Read a tested sample

Continuous sampling means testing separate small samples at fixed intervals while the reaction continues. Blue-black means starch detected; orange-brown remaining means starch not detected in that drop.

Keep the comparison valid

Change pH; measure endpoint time / s. Match temperature, enzyme and starch amounts, volumes and the endpoint rule. Prepare labelled iodine wells before mixing; keep iodine out of the main reaction.

State the timing limit

The 30 s interval gives a window, not an exact midpoint. If the 60 s sample were missing, 30 s blue-black and 90 s orange-brown would give a 60 s window. Preserve a missed sample or uncertain colour.

Use: Amylase and pH: Continuous-Sampling Investigation

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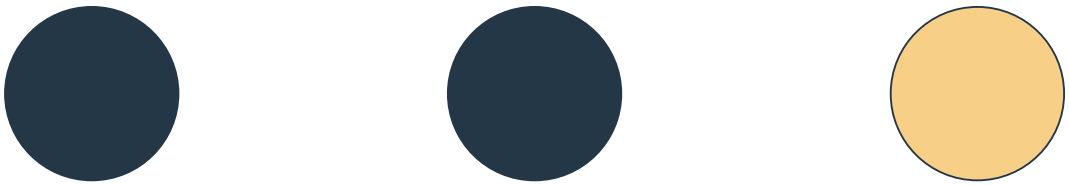
Knowledge organiser · continued

Use a bounded conclusion

First-series endpoint times: pH 4 = 240 s; 5 = 180 s; 6 = 120 s; 7 = 90 s; 8 = 150 s. pH 7 is fastest in this illustrative series. One trial per pH does not prove an exact universal optimum.

Diagram

Dark blue shows blue-black iodine. Orange shows iodine that stays orange-brown. The words give the colour result too.



30 s	60 s	90 s
Blue-black	Blue-black	Orange-brown

Supplied example: samples every 30 s.

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

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Word page

Enzyme and substrate

enzyme — A biological catalyst. It speeds up a reaction. It is not used up.

substrate — The substance that an enzyme acts on.

active site — The part of an enzyme where the substrate binds. The shapes fit each other.

specificity — An enzyme acts on particular substrates. They fit its active site.

denaturation — A change in an enzyme's three-dimensional shape. It disrupts the active site and how it works. An unsuitable pH or high temperature can cause this.

Measure and compare

pH — A measure of how acid or alkaline a substance is.

independent variable — The factor we choose to change. Here we change buffer pH.

dependent variable — The outcome we measure. Here it is endpoint time in seconds.

control variable — A factor we keep the same. This makes the comparison fair. Here we test the effect of pH.

endpoint — The first sampled time when iodine stays orange-brown.

buffer — A solution that helps keep the chosen pH.

repeat — A further trial. Keep the same stated conditions.

Use: Amylase and pH: Continuous-Sampling Investigation

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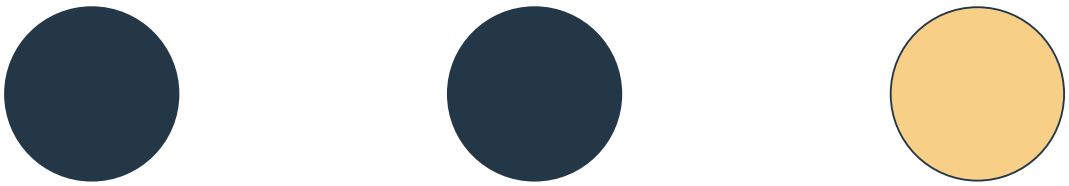
Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Word page · continued

Diagram

Dark blue shows blue-black iodine. Orange shows iodine that stays orange-brown. The words give the colour result too.

Continuous sampling means testing separate small samples at fixed intervals while the reaction continues. Blue-black means starch detected; orange-brown remaining means starch not detected in that drop.



30 s	60 s	90 s
Blue-black	Blue-black	Orange-brown

Supplied example: samples every 30 s.

Use: Amylase and pH: Continuous-Sampling Investigation

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Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

We Do · response cards

Response cards

A B C

Guided example

In your book: Complete the shared task; record one decision and its reason.

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

We Do · cards and recording

Use the cards

In your book: Complete the shared task; record one decision and its reason.

Choose a card, then a destination. You can change your mind.

Card 1

Ready tile: Goggles are on, iodine wells and labels are ready, roles are agreed and the teacher has authorised the setup.

Card 2

Iodine spill: Iodine is spilled on the bench or skin.

Card 3

Unlabelled buffer: A buffer container has no clear pH label.

Card 4

Missed interval: The 60 s sample was missed but the reaction and later observations are otherwise intact.

Card 5

Borderline colour: A well looks purple-brown and the agreed endpoint cannot be judged confidently.

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

We Do · cards and recording · continued

Card 6

Clean fixed sample: A clean transfer is made at the agreed interval to the next labelled iodine well.

Card 7

Temperature outside range: You check the temperature. It is outside the allowed range in the centre method.

Destinations

CONTINUE AS AUTHORISED

STOP AND CHECK ADULT

RECORD, THEN REPEAT IF AUTHORISED

In your book: Complete the shared task; record one decision and its reason.

Guided example

In your book: Record one evidence claim and explain its limit.